Summative Assessment RM&DA

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Short Brief

- This is an open-book exam. You may use any resources available to you (notes, books, online documentation).
- You are expected to work individually. Collaboration is not permitted.

- All code and answers must be clearly presented in this document.
- Ensure this document renders successfully to both an HTML and a PDF file.
- Show all your work, including the R code used to arrive at your answers.

Section 1: Data Wrangling

This section uses the `dataset`, which contains data about various car models.

Step 1 - Load the database

Load the dataset

```
# Place your code here
```

Inspect the dataset for basic characteristics such as type of variables, dimensions of the dataset, etc.

```
# Place your code here
```

Step 2 - Data wrangling

- Create a new variable based on an existing one

The `am` column in the `mtcars` dataset represents the transmission type (0 = automatic, 1 = manual). Create a new factor variable called `transmission_type` based on the am column to acheive the result below:

```
                  am transmission_type
Mazda RX4          1            Manual
Mazda RX4 Wag      1            Manual
Datsun 710         1            Manual
Hornet 4 Drive     0         Automatic
Hornet Sportabout  0         Automatic
Valiant            0         Automatic
```

- Create transmission_type

```
# Place your code here
```

Step 3 - Data Summarisation

Calculate the mean mpg (miles per gallon) for cars with 4 cylinders (cyl) and for cars with 6 cylinders.

- Calculate mean `mpg` for 4 and 6 cylinder cars

```
# A tibble: 2 x 2
    cyl mean_mpg
  <dbl>    <dbl>
1     4     26.7
2     6     19.7
```

```
# Place your code here
```

Question 4 ([Insert points here])

Filter the data to include only observations where the car has a manual transmission and has more than 4 gears. How many observations meet these criteria?

manual_and_more_than_4_gears <- mtcars %>% filter(am == 1 & gear > 4) nrow(manual_and_more_than_4_gears)

Your answer here

Section 2: Data Visualization This section continues to use the mtcars dataset.

Question 5 ([Insert Points Here])

Create a scatter plot showing the relationship between mpg and wt (weight). Add appropriate labels for the axes and a title for the plot.

Create scatter plot of mpg vs wt

ggplot(mtcars, aes(x = wt, y = mpg)) + geom_point() + labs(title = "Relationship between MPG and Weight", x = "Weight (1000 lbs)", y = "MPG")

Your written answer for Question 5 goes here.

Question 6 ([Insert Points Here])

Generate a boxplot to visualize the distribution of mpg across different cyl (number of cylinders) values.

Create boxplot of mpg by cyl

ggplot(mtcars, aes(x = factor(cyl), y = mpg)) + geom_boxplot() + labs(title = “MPG Distribution by Number of Cylinders”, x = “Number of Cylinders”, y = “MPG”)

Your written answer for Question 6 goes here.

Question 7 ([Insert Points Here])

Create a bar chart showing the count of cars for each number of cylinders (cyl).

Create bar chart of cylinder counts

ggplot(mtcars, aes(x = factor(cyl))) + geom_bar() + labs(title = “Car Count by Number of Cylinders”, x = “Number of Cylinders”, y = “Count”)

Your written answer for Question 7 goes here.

Section 3: Basic Statistics This section uses the mtcars dataset.

Question 8 ([Insert Points Here])

Perform an independent samples t-test to compare the mean mpg for cars with automatic and manual transmissions. State the null and alternative hypotheses, report the p-value, and interpret the results.

Perform t-test

t.test(mpg ~ am, data = mtcars)

Your written answer for Question 8 goes here.

Question 9 ([Insert Points Here])

Calculate the Pearson correlation coefficient between mpg and hp (horsepower). Interpret the strength and direction of the correlation.

Calculate correlation

cor(mtcars$mpg, mtcars$hp)

Your written answer for Question 9 goes here.

Question 10 ([Insert Points Here])

Conduct a chi-squared test to examine the relationship between cyl (number of cylinders) and am (transmission type). State the null and alternative hypotheses, report the chi-squared statistic and p-value, and interpret the results.

Perform chi-squared test

table_cyl_am <- table(mtcars$cyl, mtcars$am) chisq.test(table_cyl_am)

Your written answer for Question 10 goes here.

Question 11 ([Insert Points Here])

Fit a linear model to predict mpg using wt (weight) and hp (horsepower). Provide the R code, summarize the model, and interpret the coefficients for wt and hp.

Fit linear model

linear_model_mpg <- lm(mpg ~ wt + hp, data = mtcars) summary(linear_model_mpg)

Your written answer for Question 11 goes here.

Question 12 ([Insert Points Here])

Fit a logistic regression model to predict am (transmission type) using mpg and wt (weight). Provide the R code, summarize the model, and interpret the odds ratio for mpg.

Fit logistic model

logistic_model_am <- glm(am ~ mpg + wt, data = mtcars, family = "binomial") summary(logistic_model_am) exp(coef(logistic_model_am))

Your written answer for Question 12 goes here.

Section 4: Basic Introduction to Multivariate Analysis This section uses the iris dataset.

Question 13 ([Insert Points Here])

Briefly explain the purpose of Principal Component Analysis (PCA). In the context of the ‘iris